结构体排序

人员

臧晨云 隋淏琛 于昊华 初星熠 霍佳铭 张高 陈之恒 何佩瑜 刘奕男 刘子郴 到课

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/4950

王向东老师周六一点半C++自定义函数

刷新

#	用户名	姓名	编程分	时间	A	B	C	D	E	F	G	H	I
1	suihaochen	隋淏琛	900	9363	100	100	100	100	100	100	100	100	100
2	zangchenyun	臧晨云	900	9500	100	100	100	100	100	100	100	100	100
3	huojiaming	霍佳铭	900	10061	100	100	100	100	100	100	100	100	100
4	liuyinan	刘奕男	800	8143	100	100	100	100	100	100	100	100	
5	zhanggao	张高	800	9341	100	100	100	100	100	100	100	100	
6	chuxingyi	初星熠	800	9528	100	100	100	100	100	100	100	100	
7	hepeiyu	何佩瑜	800	10243	100	100	100	100	100	100	100	100	
8	liuzichen123	刘子郴	700	10049	100	100	100	100	100	100	100		
9	chenzhiheng	陈之恒	700	10324	100	100	100	100	100	100	100		
10	lijingxuan	李景翾	600	9636	100	100	100	100	100	100	0		
11	yuhaohua	于昊华	600	9697	100	100	100	100	100	100	0		

作业

https://cppoj.kids123code.com/contest/5142 (作业是 A ~ F 题必做)

课堂表现

今天课上讲了 结构体 和 结构体排序 的知识, 这部分知识不难, 同学们课上掌握不错, 但是还不熟, 需要课下再复习复习。

课堂内容

成绩排序

```
#include <bits/stdc++.h>

using namespace std;

struct node {
  string s;
  int c;
} a[25];

bool cmp(node p, node q) {
```

```
  if (p.c != q.c) return p.c > q.c;
  return p.s < q.s;
}

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; i++) cin >> a[i].s >> a[i].c;
  sort(a+1, a+n+1, cmp);
  for (int i = 1; i <= n; ++i) {
    cout << a[i].s << " " << a[i].c << endl;
  }
  return 0;
}
```

坐标排序

```
#include <bits/stdc++.h>

using namespace std;

struct node {
  int x, y;
} a[10005];

bool cmp(node p, node q) {
  if (p.x != q.x) return p.x < q.x;
  return p.y < q.y;
}

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; i++) cin >> a[i].x >> a[i].y;
  sort(a+1, a+n+1, cmp);
  for (int i = 1; i <= n; ++i) {
    cout << a[i].x << " " << a[i].y << endl;
  }
  return 0;
}
```

期末考试成绩排名

```
#include <bits/stdc++.h>

using namespace std;
```

```
struct node {
  int id;
  string s;
  int c;
} a[105];

bool cmp(node p, node q) {
  if (p.c != q.c) return p.c > q.c;
  return p.id < q.id;
}

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; i++) cin >> a[i].id >> a[i].s >> a[i].c;
  sort(a+1, a+n+1, cmp);
  for (int i = 1; i <= n; ++i) {
    cout << a[i].id << " " << a[i].s << " " << a[i].c << endl;
  }
  return 0;
}
```

生日

```
#include <bits/stdc++.h>

using namespace std;

struct node {
  string s;
  int y, m, d;
  int id;
} a[105];

bool cmp(node p, node q) {
  if (p.y != q.y) return p.y < q.y;
  if (p.m != q.m) return p.m < q.m;
  if (p.d != q.d) return p.d < q.d;
  return p.id > q.id;
}

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; i++) {
    cin >> a[i].s >> a[i].y >> a[i].m >> a[i].d;
    a[i].id = i;
  }
```

```
  sort(a+1, a+n+1, cmp);
  for (int i = 1; i <= n; ++i) {
    cout << a[i].s << endl;
  }
  return 0;
}
```

宇宙总统

```
#include <bits/stdc++.h>

using namespace std;

struct node {
  int id;
  string c;
} a[25];

bool cmp(node p, node q) {
  if (p.c.size() != q.c.size()) return p.c.size() > q.c.size();
  return p.c > q.c;
}

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; i++) {
    cin >> a[i].c;
    a[i].id = i;
  }
  sort(a+1, a+n+1, cmp);
  cout << a[1].id << endl;
  cout << a[1].c << endl;
  return 0;
}
```

合影

```
#include <bits/stdc++.h>

using namespace std;

struct node {
  string s;
  int c;
} a[45];

```

```cpp
bool cmp(node p, node q) {
  if (p.s=="male" && q.s=="male") {
    return p.c < q.c;
  } else if (p.s=="male" && q.s=="female") {
    return true;
  } else if (p.s=="female" && q.s=="male") {
    return false;
  } else {
    return p.c > q.c;
  }
}

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; i++) {
    cin >> a[i].s >> a[i].c;
  }
  sort(a+1, a+n+1, cmp);
  for (int i = 1; i <= n; ++i) cout << a[i].c << " ";
  cout << endl;
  return 0;
}
```